

Exploring Interaction, Movement and Video Game Experience in an Educational VR Experience

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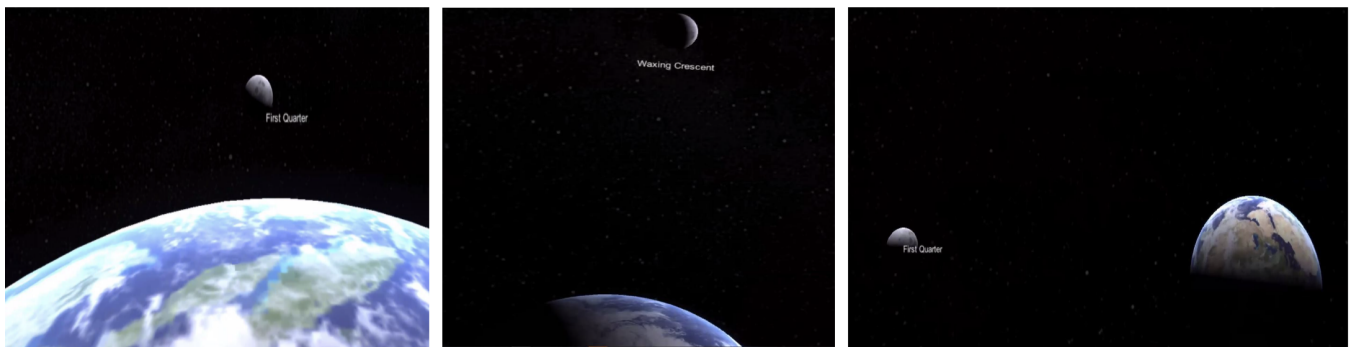


Figure 1: Learners could view the Moon from three positions: Surface (left), North Pole (middle), and Far Above (right).

ABSTRACT

Findings of increased engagement and preference for XR learning are well-established. However, there are concerns that the high sense of presence that virtual reality can engender may increase cognitive load and actually decrement learning outcomes [16]. Recent research has highlighted the importance of understanding the contributions of embodiment to learning in virtual reality [13]. In this paper, we analyze previously unpublished secondary data of participant interactions, movements, and presence ratings to investigate how user behavior may predict learning in embodied, immersive virtual environments, and how this can guide the development of broadly useful XR interventions. We find that increased global movement predicts higher post-test scores, but more embodied interactions (i.e., dragging the Moon using hand gestures rather than button presses) do not improve learning. Rather, actions linked to greater familiarity with video game play predict greater learning of the subject matter.

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CCS CONCEPTS

• **Human-centered computing** → **Human computer interaction (HCI); Interaction paradigms; Virtual reality;**

KEYWORDS

Virtual Reality, Learning, Presence, Embodiment, Movement

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1 INTRODUCTION

Immersive virtual reality (VR) applications have been widely developed for education [15, 17] with expected positive effects ranging from improving learning outcomes to motivating user interest [8]. Findings of increased engagement and the preference for VR learning have been well established [3, 9] even when they do not align with improved learning [14]. This disconnect has led some to argue that while VR's compelling visual effects and highly immersive simulations can enhance learners' sense of presence and engage their interest, too much visual information and complex interactivity could overload learners and harm learning outcomes [2]. Others have argued that properly designing learning experiences to consider embodiment and presence can unlock the benefits of virtual

reality [13]. Below, we briefly review the literature on embodiment, presence, and the effects of prior video game experience on learning in immersive virtual reality. We then present some exploratory findings on learners' behaviors as they were recorded during a VR learning experience, and how these behaviors were related to learning outcomes, presence measures, and previous video game experience.

Students frequently name a sense of presence as a reason to prefer VR as a learning medium [14, 16]. However, previous work indicates that feeling “present” in VR can be a double-edged sword. As Huang et al. [11] found, “regression coefficients of involvement/immersion ... positively predicted learning satisfaction”, but negatively predicted performance on a subsequent knowledge test. Albus et al. [2] suggest that virtual reality may increase cognitive load, such that higher presence is linked with reduced learning compared to less immersive methods. This aligns with findings from Bailey et al. [5], showing that transferring a desktop simulation to immersive virtual reality increased presence, yet reduced learning, a deficit attributed to overload. In a recent study, Ahn and colleagues [1] describe the unintended consequences of higher presence, which decreased recall in participants learning about ocean acidification.

However, Johnson-Glenberg et al. [13] suggest that the benefits of embodied cognition outweigh this concern. Specifically, they suggest that virtual reality experiences that properly implement embodiment can improve learning over non-immersive experiences, beyond simply increasing engagement. For example, vocabulary learning is improved by object manipulation in virtual reality [10]. Similarly, compared to passive viewing, directly manipulating the object achieves better learning outcomes in anatomy training [12]. Johnson and colleagues described presence as one of the three key components of embodiment, the other two being the congruence of the gesture to the content, and the magnitude of the gesture. In their study crossing platform immersivity (2D or 3D experiences) or level of embodiment, the authors found that “high embodiment” conditions outperformed “low-embodiment” conditions even when less-immersive platforms were used. Their paper highlights the importance of designing for the platform and argues that not all topics can or should be taught in immersive virtual reality.

Many past studies found that body movement plays a unique role in the learning process. Embodied interactions can engage learners, concretize abstract concepts and facilitate the learning process [27]. However, such embodied interactions must be effective. As Cafaro et al. [7] found out, the “uncanny valley” also exists when mapping learners' movements to corresponding virtual effects. When mapping is appropriate, learners can explore the virtual environment with higher engagement. Bermejo-Berros and Gil Martínez [6] detected different types of exploration behaviors. The first type described learners who moved their heads quickly to acclimate to the environment without focusing on particular objects. The second type of learners then examined the environment carefully after an initial exploration [6]. This leads to our first research question: **RQ1: What is the relationship between tracked participant behavior (event and movement data) and learning outcomes?**

To more granularly examine the relationship between presence, behavior, and learning outcomes in virtual reality, we use a previously unanalyzed dataset from a study in which participants learned

about the phases of the Moon in an immersive virtual reality experience [14]. In this study, participants could manipulate the phases of the Moon by moving it through its phases and observing how its appearance changed as its relationship with the Earth and the Sun changed. The experience in the VR simulation is similar to the common classroom demonstration in which a ball on a stick is held at arm's length as the student rotates in front of a light source [19]. There were no differences in learning between VR and non-VR conditions. In the VR version, learners could either use a “more embodied” method to move the Moon by dragging it or a “less embodied” method by pressing triggers and grips. These different embodied interactions allow us to set up the second research question: **RQ2: Are behaviors that might indicate a greater experience of embodiment (in this case, the types of gestures used to engage with the virtual content) linked to greater presence?**

In the study from which these data were derived, researchers reported that participants with experience with video games attained higher scores than participants without video game experience, but the data from participants in the virtual reality condition was not further explored in that paper. Work by Madden et al. [14] and Smith et al. [21], has suggested that differences in learning in immersive VR might be accounted for by familiarity with related content, such as video games, and that scaffolding learning in virtual reality through providing equivalent experiences could provide even inexperienced participants with optimal learning experiences. This aligns with research suggesting that learners' previous experiences in video games could transfer to other VR experiences. For example, Schlickum et al. [20] found that “... experience of video games are important for training outcome in simulated surgical procedures. Video game training can become useful when designing future skills training curricula for surgeons.” This leads us to our third research question: **RQ3: How might participant behaviors and presence ratings be linked to previous experience with video games?**

2 METHODS AND MEASURES

In this paper, we analyzed previously un-examined movement data and events, as well as self-reported presence measures, from participants who experienced a virtual reality Moon Phases learning experience.

In this experience, participants were assisted in donning an Oculus Rift and shown how to use hand controllers, and were given instructions by an RA on how to navigate a virtual environment portraying the solar system. In this environment, participants could move a model of the moon through its phases to see how the phases of the moon change over time. They could do so either by “grabbing” and “dragging” the Moon using a laser pointer from the controller (the more embodied method of interaction) or by pressing buttons on the right hand controller (the less embodied method of interaction). On the left-hand controller, participants saw a menu that presented a series of 10 multiple-choice questions about Moon phases, which they were allowed to answer multiple times. Participants could jump to three different positions in the universe to view the Moon, Earth and Sun from different perspectives.

The X, Y, and Z positions of participants' head were recorded for the entire session. In-world events were also tracked and recorded; for example, whether participants were moving the Moon by dragging or by button-press. None of these measures have previously been reported.

2.1 Participants

Of 51 participants, one participant's data was removed due to an extremely long duration of about 12 hours indicating a flaw in the recording process, leaving 50 participants (42 females, 7 males and 1 other). There were 49 participants in the age range of 18 - 24 and 1 participant aged 45+. Participants self-reported race and ethnicity as African American (6), Asian/Pacific Islander (24), Caucasian (17), and Hispanic or Latino (3), and 2 participants preferred not to specify. All procedures were approved by the university IRB, and all participants signed informed consent.

2.2 Measures

The measures of the study include the data tracked within and outside the virtual environment. During the virtual experience, participants' movement data as well as events that characterized their interactions with the VR environment were tracked. After participants' virtual experiences, they completed a self-report survey and post-test. Additional measures include the results from self-report survey questions on presence and learning. We described all the events recorded and the measures captured from the post-experiment survey below.

2.2.1 Events data. The following events and the time stamps at which they occurred were recorded during the experiment:

Dragging the Moon: Participants could use the laser from the hand controller to select and then drag the Moon through its orbit using large arm movements. This is a more embodied way to move the Moon. The percentage of time in seconds participants dragged the Moon out of the total time they interacted with the Moon ($M = 0.213$, $SD = 0.247$) and the percentage of time in seconds participants dragged the Moon out of the total time they spent in the VR experience ($M = 0.031$, $SD = 0.033$) were reported.

Moving the Moon through trigger or trigger and grip presses:

Participants could also use the trigger to advance the Moon through its phases, which could be accelerated by pressing the trigger and the grip buttons together for a faster speed. This is a less embodied way to move the Moon. The percentage of time in seconds participants either used "trigger" or "trigger and grip" to move the Moon out of the total time they interacted with the Moon ($M = 0.821$, $SD = 0.239$) and the percentage of time in seconds participants either used "trigger" or "trigger and grip" to move the Moon out of the total time they spent in the VR experience ($M = 0.179$, $SD = 0.094$) were reported.

Jumping to a different position: When participants decided to jump from one position to another position, a "jump" event along with one of the three positions they jumped to were recorded: North Pole position, Surface position and Far Above position (Figure 1). The number of jumps participants made from position to position throughout the VR session was reported ($M = 6.400$, $SD = 5.500$). Not all participants jumped; some participants remained in the starting position at the North Pole for the entire interaction.

2.2.2 Movement. Movement can be characterized in two ways: global and local movements. We define global movement as the sum of head position changes throughout the experiment, calculated as the Euclidean distance derived from the positional data (X, Y, Z coordinates) of the headset from one time point to the next. The positional data reflected the coordinates in the universe, which were based on the radius of the Earth. This meant that participant movements in the physical world did not always map closely to their movement through the virtual universe. For example, when participants changed their position from being embedded in the surface of the Earth position to jumping to hovering far above it, their position in the physical world changed little if at all. Similarly, participants' position "in the universe" would also change when they moved the Moon, since this meant that time was passing and the Earth would also travel around the Sun. Thus, this type of movement could be characterized as a measure of exploration of the virtual environment ($M = 72723.72$, $SD = 30476.740$).

Local movement refers to how the movement of participants' head relates to behavior in the physical world [28]. For example, local movement might indicate postures that might indicate emotional states or personality traits [22], or scanning behavior that reflects states of mind [26]. Such movement data can potentially be used to re-identify participants [18] or reveal individual characteristics [28] or even physical conditions of participants [4]. In this study, while all participants started in the North Pole position, not all participants chose to move to Surface and Far Above positions. This created challenges when comparing movement in the different positions, since the scale of participants' virtual body also changed to allow them to interact with the Moon from each viewpoint. Therefore, we created "local" movement measures using only head movement data in the North Pole position, during the period when there were no events triggered (when the Moon was not being moved by the participant). Similarly, the Euclidean distance of the movement between the timestamps was summed using the head X, Y, Z positional data. This allowed us to use consistent movement measures for individual level results to characterize "local" movements that directly reflected participants' head movements in the physical world ($M = 23255.690$, $SD = 26752.080$).

2.2.3 Presence. We used two measures of presence. One, based on Witmer and Singer [24] captured participants' reactions to the immersive qualities of the virtual environment. Following the factor structure of the survey [23], those 10 presence questions were broken into 5 categories on a 5-point scale. We calculated internal consistency across these questions for each of five categories (when relevant; some were one-item scales), and we used the "adaptation" category because the participants' responses to these questions were internally consistent ($\alpha = 0.72$, $M = 3.965$, $SD = 0.664$).

The second set of presence questions are environmental presence questions. These questions were modified from Won et al. [25] to capture the sense of "being there" ($\alpha = 0.73$, $M = 3.198$, $SD = 0.738$). Both sets of survey questions are listed in Appendix.

2.2.4 Learning outcomes. Participants first took a pre-test on their knowledge of Moon phases ($M = 37.890\%$, $SD = 17.957\%$) and then took a post-test after the VR experience ($M = 59.504\%$, $SD = 19.651\%$).

2.2.5 Experience with video games. Following Madden et al. [14], we examined how these measures differ among participants with different levels of previous video game experience. No participants responded that they played video games daily. Six participants reported playing weekly; six participants reported playing 1-2 times a month; 20 participants reported playing 1-2 times a year; and 16 participants answered “never”; two participants did not answer the survey question. The twelve participants who reported “daily”, “weekly”, or “1-2 times a month” video game play were categorized as “having significant video game experience” and the 36 participants who reported “1-2 times a year” or “never” were categorized as “not having significant video game experience”.

3 RESULTS

All analyses were conducted using R. The code package can be found here (<https://doi.org/10.6077/p2a6-dj12>).

3.1 RQ1: Do participant behaviors (movement or actions) or presence predict learning outcomes?

We began with a linear model using the `lm` package in R. Based on the correlation matrix results in Appendix, we included the percentage of time participants pressed the trigger and grip or only the trigger over the time spent in the experience, the percentage of time participants spent dragging the Moon over the time spent in the experience, the average of their adaptation score, the total duration of participants’ time in VR, their environmental presence score and the pre-test score as predictors. Controlling for the environmental presence score, their adaptation score, and pre-test scores, we found participants’ interactions with the virtual Moon using the trigger and grip positively predicted their post-test scores. As participants moved the virtual Moon more using the trigger and/or grip ($F(1,33) = 7.309, p = 0.011, 95\% \text{ CI } [2.378e-01, 1.684]$), their post-test scores also improved. In other words, the more time users spent in the less-embodied interaction, moving the Moon by trigger and/or grip-press, correlated with higher post-test scores. Similarly, the pre-test scores marginally predicted the post-test scores ($F(1,33) = 3.623, p = 0.066, 95\% \text{ CI } [-2.107e-02, 0.633]$). These results remained the same whether or not presence and adaptation were controlled for in the model. In addition, the relationship between total time in VR and post-test scores was marginally statistically significant ($F(1,33) = 3.648, p = 0.065, 95\% \text{ CI } [-1.556e-05, 4.931e-04]$).

However, the percentage of time that participants dragged the Moon did not predict higher post-test scores ($F(1,33) = 0.444, p = 0.510, 95\% \text{ CI } [-1.483, 2.927]$). In other words, spending more time with the interaction we considered to be more naturalistically embodied, in which users moved their arms in large arcs to drag the Moon across the sky matching their gestures in the physical world, did not improve learning.

The adaptation component of Witmer-Singer’s presence measures predicted post-test scores: ($F(1,33) = 7.533, p = 0.010, 95\% \text{ CI } [2.975e-02, 0.200]$). As adaptation scores rose, so did post-test scores. However, the relationship between environmental presence scores and post-test scores was not statistically significant ($p > .25$).

Examining the relationship between global movement and the post-test scores while controlling for the pre-test scores, we found

that the global movement did predict post-test scores ($F(1,47) = 7.144, p = 0.010, 95\% \text{ CI } [5.147e-07, 3.647e-06]$). We then examined local movement, which only used the head movements of participants in the North Pole position because all participants had movement measures for this position. And this allowed us to avoid comparing other positions that might have prompted different conditions (see Table 2 in Appendix, which shows how interactions differed depending on whether participants were in the Surface, North Pole, or Far Above positions). Using a linear regression model and again controlling for highly predictive pre-test scores, we see that the amount that participants moved around in the physical space during the experience does *not* predict post-test scores, ($F(1,47) = 0.239, p = 0.627, 95\% \text{ CI } [-1.487e-06, 2.441e-06]$).

3.2 RQ2: Are behaviors that might indicate a greater experience of embodiment (i.e., tracked participant behavior) linked to greater presence?

To answer this question, we examined interactions and movements separately.

RQ2a: Did participants’ interactions predict their sense of presence? The behaviors of dragging the Moon or using the grip and the trigger to move the Moon were not predictive of adaptation scores or environmental presence scores (all p ’s > 0.1). The number of times participants jumped from one position to another did predict their adaptation score ($F(1,36) = 5.088, p = 0.030, 95\% \text{ CI } [0.005, 0.092]$). However, we cannot conclude whether the experience of jumping increased participants’ feelings of adaptation, or if participants who adapted faster simply explored more.

RQ2b: Did participants’ movement predict their sense of presence? There were no statistically significant relationships between local or global movement and environmental presence or adaptation scores (all p ’s > 0.1).

3.3 RQ3: How might participant behaviors and presence ratings be linked to previous experience with video games?

We found a number of differences between experienced and inexperienced participants (see Table 1 for an overview). However, the areas in which these two groups did *not* differ were also important; for example, the *pre-test* scores of participants with more or less video game experience did not differ significantly.

There were significant differences in participants’ average adaptation scores, a subset of the Witmer and Singer presence measures ($t(36.369) = -2.348, p = 0.024$), such that participants with more video game experience reported a higher degree of adaptation to the virtual environment ($M = 4.250, SD = 0.379$) than did participants with less previous video games experience ($M = 3.870, SD = 0.714$). However, there were no significant differences in participants’ average environmental presence scores ($t(15.638) = -0.147, p = 0.885$). Summary statistics for the subsets of the presence questions averaged for groups of participants with less or more video game experience are listed in Table 3 in Appendix.

Participants with more video game experience jumped to a different position more often than participants with less video game

experience ($t(16.752) = -2.580$, $p = 0.020$). Participants with more previous video game experience also spent a marginally significant higher percentage of their total time in the experience dragging the Moon over the entire time they spent in the VR experience (a more embodied type of interaction) ($M = 0.053$, $SD = 0.052$) compared to those with less previous video game experience ($M = 0.024$, $SD = 0.019$) ($t(11.044) = -1.797$, $p = 0.100$). There was no significant difference between the two groups in the percentage of time they spent using the trigger or the trigger and the grip to advance the Moon phases ($t(16.022) = -0.201$, $p = 0.843$). However, because both groups of participants spent much more time using the trigger and grip, there was no significant difference in the *proportion* of time dragging the Moon between participants with less video game experience ($M = 0.170$, $SD = 0.165$) or more video game experience ($M = 0.329$, $SD = 0.387$), ($t(11.415) = -1.314$, $p = 0.215$).

There were also no significant differences between participants' local movements ($t(32.760) = -0.541$, $p = 0.592$) regardless of their previous video game experience. However, there was a significant difference in global movement ($t(20.018) = -2.196$, $p = 0.040$) between participants with more video game experience ($M = 89305.700$, $SD = 28252.580$), who explored the virtual environment more, and less video game experience ($M = 68276.150$, $SD = 30131.020$).

4 DISCUSSION

In this study, we examined the behavioral and self-report data from participants who experienced a virtual reality environment designed to teach about the phases of the Moon. We found that global movement, defined as exploratory movement throughout the virtual environment that reflected multiple means of exploration, predicted learning gains. However, the physical movements of participants alone were not predictive of learning.

Though it is a less embodied way to interact with the Moon in the virtual environment, participants who used the trigger and/or the trigger and grip more got higher scores on the post-test.

4.1 Limitations

Our primary limitation is that this data set is small and the findings are exploratory. While we have reported all of the tests we conducted in this document and the appendices, the explanations we have proposed still need to be tested. For example, we did not see that participants' theoretically more-embodied action of dragging the Moon predicted better learning outcomes. However, as participants moved to different positions, their ability to interact with the Moon also differed. For example, in the North Pole position, participants could point to the Moon more easily using their controller's laser pointer than on the Surface and Far Above positions. Additional experiments explicitly designed to cleanly compare different types of embodied conditions are thus needed. There are other potential types of embodied actions that might be more relevant; for example, touching the Moon with a hand controller, or grasping the Moon using hand tracking; and the role of avatar embodiment might also have important effects.

4.2 Design Implications

Although these results are exploratory, we propose they offer some insights for designing educational virtual reality experiences. In

addition to addressing our initial research questions, our analysis demonstrates how VR projects can provide rich data collection to support VR developers, designers, and educators.

Using behavioral data, people can drill down on learners' behavior at the question level. By answering such questions as "Did participants jump from one position to the other when they answer the question?", "Did participants' attention follow the Moon as they changed the position of the Moon?", such derived measures can lead to insights to help the educators to understand not only "how" learners investigate and resolved questions in VR, but also "why" they made decisions to investigate in a certain way.

In order to more fairly test the effects of embodiment, future UX design could consider enhancing the intuitiveness of an embodied process like dragging the Moon. For example, a visible targeted laser point projected directly on the object could potentially help users to understand how to "drag" the Moon and make this interaction more accessible. This might make the interaction more comparable to traditional methods of interaction such as pressing buttons.

5 CONCLUSION

In this paper, we provide evidence to suggest that the ability to navigate through virtual space—not the level of embodiment of the method used to do so—predicts increased learning in virtual spaces by promoting exploration. We propose that this exploratory behavior is, in turn, predicted by participants' experience with similar systems; for example, video game play. This aligns with previous work showing that video game experience helped participants learn from VR; our results provide a possible explanation for why this might be so. Further, we argue that participants will often prefer to use the most efficient and familiar method of movement, rather than the most naturalistic; for example, moving objects using button presses rather than a more natural and more embodied "grip and drag" movement.

Specifically, we propose that the patterns found in this data set show two types of participant behavior related to video game experience. The "experienced" participant, who reports video game experience frequency of "weekly", and "1-2 times a month" uses button presses and triggers to move the Moon, and jumps around more from location to location. They have higher adaptation scores, although they do not report a greater sense of presence in other measures. This behavior aligns with exploratory behavior, in which transforming physical behavior leads to greater abilities in the virtual world. These participants score better on the post-test, although their pre-test scores as a group are the same on average as non-experienced participants. In contrast, inexperienced participants, while they spend the same amount of time in the virtual world and interact with the Moon using the trigger press and grip, do not adapt to, and do not explore, the virtual environment, and thus do not improve their understanding of the subject matter.

We propose that future work endeavoring to unlock the capabilities of virtual reality should focus on encouraging and facilitating exploratory behavior, and should consider how participants' previous experiences interact with embodied experiences to promote such behavior.

Table 1: The average of the metrics with different levels of previous video game experience

Event Type Number of Participants	Never 16	1-2x a year 20	1-2x a month 6	Weekly 6
Pre-test score (%)	0.365	0.404	0.376	0.359
Post-test score (%)	0.558	0.567	0.687	0.685
VR test score (%)	0.520	0.572	0.75	0.693
Total duration (s)	691.725	657.595	785.483	790.476
Number of jumps	6.063	4.800	7.667	12.667
Dragging moon (s)	19.603	12.673	57.582	40.821
Moving moon with button (s)	105.377	137.907	154.921	125.438
Local head movement	30212.680	15874.070	20391.830	31813.950
Global head movement	68860.640	67808.550	83481.240	95130.170
Total number of clicks	18.375	16.950	14.833	14.333

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